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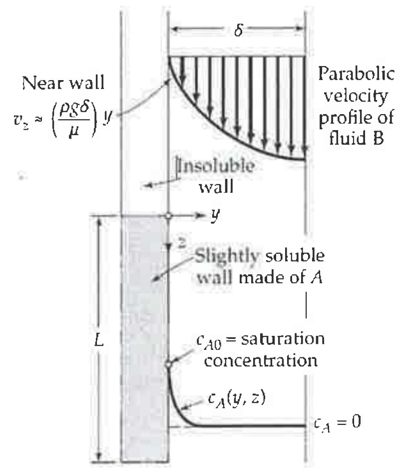
TRANSPORT PHENOMENA II, CHE 312

QUIZ 4

CLOSED BOOK, TIME: 20 MINUTES

Liquid B flows in laminar motion down a vertical wall, with the velocity profile

$$v_z(y) = \frac{\rho g \delta^2}{2\mu}\left[2\left(\frac{y}{\delta}\right) - \left(\frac{y}{\delta}\right)^2\right] \quad \left(\frac{m}{s}\right)$$



For $0 < z < L$ the wall is coated with a species A that is slightly soluble in B ($x_A \ll 1$). For short distances downstream, species A will not diffuse very far into the falling film. That is, A will be present only in a very thin boundary layer near the solid surface.

Write a shell mole balance for species A within the liquid film and derive a partial differential equation for the concentration of A as a function of y and z.

Hint: combined mole flux is $\langle N_A| = c_A\langle v^*| + \langle J_A^*|$, where the diffusive flux according to Fick's law is $\langle J_A^*| = -cD_{AB}\langle \nabla x_A|$.

$D\left(\frac{kg}{m^2 \cdot s}\right)$

$c\left(\frac{kg}{m^3}\right)$

$W = vc$

$\langle N_A| \approx \langle J_A^*|$

Shell: $\delta N_A|_z - \delta N_A|_{z+\Delta z} = 0$

$\frac{\partial N_A}{\partial z} = 0$

$\frac{d(-cD_{AB}\langle \nabla x_A|)}{\partial z} = 0$

$D_{AB}\langle \nabla x_A| \frac{dc}{dz} = 0$

B.C.

$y = 0$, $v_z \approx \left(\frac{\rho g \delta}{\mu}\right) y$

$y = \delta$, $c_A = 0$